

HEAP SORT

PROGRAM CODING:

```
#include<stdio.h>

#include<conio.h>

void heapsort(int k[],int n);

void createheap(int k[],int n);

void main()

{

    int k[25],i,n;

    clrscr();

    printf("Enter the limit:");

    scanf("%d",&n);

    printf("Enter the elements to be sorted:");

    for(i=1;i<=n;i++)

        scanf("%d",&k[i]);

    heapsort(k,n);

    getch();

}

void heapsort(int k[],int n)

{

    int q,key,i,j,temp;

    createheap(k,n);

    for(q=n;q>=2;q--)

    {

        temp=k[1];
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k[1]=k[q];
k[q]=temp;
i=1;
key=k[i];
j=1;
if((j+1)<q)
{
    if(k[j+1]>k[j])
        j++;
    while((j<=q-1)&&(k[j]>key))
    {
        temp=k[i];
        k[i]=k[j];
        k[j]=temp;
        i=j;
        j=2*i;
        if((j+1)<q)
        {
            if(k[j+1]>k[j])
                j=j+1;
            if(j>n)
            {
                j=n;
                k[i]=key;
            }
        }
    }
}

```

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    }
}

printf("\n\n The elemets after sorting:\n");

for(i=1;i<=n;i++)

printf("\t%d",k[i]);

}

void createheap(int k[],int n)

{

int i,j,q,key,temp;

for(q=2;q<=n;q++)

{

i=q;

key=k[q];

j=i/2;

while((i>1)&&(key>k[j]))

{

temp=k[i];

k[i]=k[j];

k[j]=temp;

i=j;

j=i/2;

if(j<1)

j=1;

k[i]=key;

}

}

printf("\n\n Heap generated for the elements as follows:\n");

```

```
for(q=1;q<=n;q++)  
    printf("\t%d",k[q]);  
}
```